KKR GENAI INNOVATIONS

Learn. Build. Lead With AI

FLAGSHIP PROGRAM

AI LAUNCHPAD PROGRAM

The 16-Week Generative AI Engineering Program

UNDERSTAND / BUILD / CREATE / LAUNCH

One continuous arc across **16 weeks**: master Python, **Machine Learning, Deep Learning and NLP**, engineer the prompt, look inside the **LLM**, engineer the context, supply it with RAG and tools, build the harness, control the loop, structure it as a **LangGraph**, earn your **Claude AI certification**, fine-tune on a real **RTX 5090 GPU**, and finish by shipping **2 to 4 capstone projects** that launch your career.

DURATION	CAPSTONE PROJECTS	FORMAT
16 Weeks	**2 to 4**	**Live · Zoom**

ENROLLMENT: **OPEN** · NEW BATCHES START REGULARLY

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ONE COURSE · EVERYTHING INCLUDED

How the 16 Weeks Are Structured

The program opens with the foundations every AI engineer needs: **Python, Machine Learning, Deep Learning and NLP**. From there, every discipline exists because the one before it ran out of road: you feel the prompt's ceiling before context engineering, context rot before WSCI, the single-run harness before the loop, the runaway loop before the graph. The final weeks are pure building: **your capstone projects, shipped and operated**, with a career launch alongside.

WHAT MAKES THIS PROGRAM

ML, DL & NLP Foundations First

Three full weeks of Machine Learning, Deep Learning and NLP before a single prompt is written, so every technique that follows rests on real understanding.

The Six Disciplines

PROMPT → CONTEXT → RAG & TOOLS → HARNESS → LOOP → GRAPH: six disciplines in sequence, each introduced because the one before it ran out of road.

Projects at the End

Weeks 14 to 16 are dedicated to capstones: an Enterprise Knowledge Assistant in production RAG and a multi-agent Customer Support Agent, with scope for four.

Claude AI Certification

A full certification week on the Claude ecosystem: Projects, Artifacts, Claude Code and the Messages API, ending in a graded build project and assessment.

Real GPU Fine-Tuning

LoRA and PEFT fine-tuning runs on the NVIDIA RTX 5090 machine in our Alpharetta GPU lab: applied deep learning on real hardware, never on a simulator.

Career Launch Built In

Cloud deployments on Azure OpenAI, AWS Bedrock and Vertex AI, a 30-project GitHub portfolio, an AI résumé, interview preparation and your own Demo Day.

THE COMPLETE ARC

16 Weeks at a Glance

01	**Foundations: Python for AI Engineers**	Days 1–7	Python for AI engineers, from zero to your first LLM call
02	**Machine Learning Fundamentals**	Days 8–14	Supervised & unsupervised learning, scikit-learn, evaluation
03	**Deep Learning & NLP Fundamentals**	Days 15–21	Neural networks, embeddings, tokenization, transformers
04	**Prompt Engineering**	Days 22–28	Shape one call, and find its ceiling
05	**LLM Fundamentals & the Claude Family**	Days 29–35	Inside the transformer; the Claude models & Messages API
06	**Context Engineering**	Days 36–42	Everything that enters the window
07	**LangChain, RAG & Tools**	Days 43–49	Retrieval & computation; the Resume Matcher build
08	**Harness Engineering**	Days 50–56	Build a working coding agent
09	**MCP: Model Context Protocol**	Days 57–63	The open standard, client to server
10	**Loop Engineering**	Days 64–70	Controlled, budgeted, self-correcting loops
11	**Graph Engineering & LangGraph**	Days 71–77	Typed state, checkpoints, human gates
12	**Claude AI Certification**	Days 78–84	Certification build project and assessment
13	**GPU Lab & Cloud Deployment**	Days 85–91	RTX 5090 fine-tuning; Azure, Bedrock & Vertex deploys
14	**Capstone I: Enterprise Knowledge Assistant**	Days 92–98	A production RAG system, designed and built
15	**Capstone II: Multi-Agent Support System**	Days 99–105	A multi-agent build, orchestrated in LangGraph
16	**Ship, Operate & Career Launch**	Days 106–112	CI/CD, evals, portfolio, Demo Day and graduation

CLASS SCHEDULE: LIVE ON ZOOM

Tue to Fri	9:00–10:00 PM EST (1 hour)	6:30–7:30 AM IST, next day
Saturday	10:00 AM–12:00 PM EST (2 hours)	7:30–9:30 PM IST

Every class is recorded on Skool · mentor 1:1 feedback · graded weekly checkpoints · the KKRGENAI Hub community on WhatsApp · every class carries a hands-on lab, and every week ends with something shipped.

Weeks 01 & 02

PHASE 1 · FOUNDATIONS

WEEK 01

YOU SHIP **Your first LLM call from your own environment**

AI Foundations: Python for AI Engineers

DAYS 1–7

The map of AI on day one, then the Python every model and agent will stand on.

- The AI landscape: ML vs DL vs NLP vs GenAI
- The stack: model → prompt → context → harness → loop → graph
- Python I–IV: syntax, control flow, functions, OOP
- Python V: generators, iterators, decorators
- Environment setup: Python, git & API keys

Stack: Python · git · API keys · your first LLM call from code

> ✓ **CHECKPOINT**
> The week closes with your environment fully working: Python written across five sessions, git and API keys configured, and your first LLM call made from your own code. You leave knowing exactly where ML, DL, NLP and GenAI sit in the stack.

WEEK 02

YOU SHIP **Your first trained and evaluated ML model**

Machine Learning Fundamentals

DAYS 8–14

The discipline underneath everything: teach a machine to learn from data.

- Supervised vs unsupervised vs reinforcement
- Regression & classification, hands on
- Train/test splits, overfitting & regularization
- Evaluation: accuracy, precision, recall
- Clustering & dimensionality reduction

Stack: scikit-learn · pandas · NumPy · an ML mini-project end to end

> ✓ **CHECKPOINT**
> You now understand how machines learn from data: framing a problem, training regression and classification models, splitting data honestly, and reading accuracy, precision and recall. Your evaluated mini-project proves you can run the full ML workflow end to end.

Weeks 03 & 04

PHASE 1 · FOUNDATIONS

WEEK 03

YOU SHIP **A neural text classifier, trained and tested by you**

Deep Learning & NLP Fundamentals

DAYS 15–21

From neurons to language: the road that leads straight to the transformer.

- Neural networks & backpropagation intuition
- A taste of PyTorch: build a small network
- Tokenization & embeddings: vectors of meaning
- NLP tasks: classification, NER, summarization
- From RNNs to attention: why transformers won

Stack: PyTorch · Hugging Face · embeddings · NLP pipelines

> ✓ **CHECKPOINT**
> Neural networks are no longer a black box: you have built one, trained it in PyTorch, and turned text into embeddings. With tokenization, classic NLP tasks and the road from RNNs to attention behind you, the transformer is ready to be understood.

WEEK 04

YOU SHIP **A production-grade system prompt with its own evaluation**

Prompt Engineering

DAYS 22–28

Shape a single call to the model, then hit the prompt's ceiling deliberately.

- System prompt engineering
- Prompt anatomy: zero / one / few-shot
- Chain-of-Thought & reasoning-time techniques
- Evaluating, optimising & attacking prompts
- Why the prompt alone is not enough

Stack: Frontier chat models · API playgrounds · prompt evals

> ✓ **CHECKPOINT**
> You can now engineer a prompt the way production teams do: structured system prompts, shot-based and reasoning techniques, and a proper evaluation loop that catches failures. You have also felt the prompt's ceiling, which is exactly why the next weeks exist.

Weeks 05 & 06

PHASE 2 · THE PROMPT & THE MODEL

WEEK 05

YOU SHIP A working Claude API app, and a transformer you can explain

LLM Fundamentals & the Claude Family

DAYS 29–35

Look inside the model you have been steering, right after learning to steer it.

- Transformer architecture I & II: attention & weights
- Training vs inference · context windows & sampling
- Why models hallucinate, and what it means
- The Claude model family & the Messages API
- Your first Claude API application, built live

Stack: Claude models · Messages API · tokens & sampling controls

> ✓ **CHECKPOINT**
> You have looked inside the model itself: attention, weights, training versus inference, context windows and sampling, and why hallucination happens. On top of that understanding sits your first working Claude application, built live with the Messages API.

WEEK 06

YOU SHIP A context-managed assistant that survives long conversations

Context Engineering

DAYS 36–42

Engineer everything that enters the window, and manage it before it rots.

- Context engineering as a discipline
- The elements of context
- RAG & tools as context supply strategies
- Context rot: why more context is worse
- WSCI: Write, Select, Compress, Isolate

Stack: Context assembly patterns · long-conversation memory strategies

> ✓ **CHECKPOINT**
> Everything that enters the window is now under your control: the elements of context, RAG and tools as supply strategies, context rot, and the WSCI discipline. Your assistant survives long conversations because you engineered its memory, not because you got lucky.

Weeks 07 & 08

PHASE 3 · CONTEXT, RAG & THE HARNESS

WEEK 07

YOU SHIP **The Resume Matcher: your first shipped RAG application**

LangChain, RAG & Tools

DAYS 43–49

The two mechanisms that fill the window: retrieval and computation.

- LangChain foundations & structured output
- RAG I: embeddings & the indexing side
- RAG II: the retrieval side
- Agents & tools primer
- Guided build: the AI Resume Matcher

Stack: LangChain · vector stores · embeddings · tool calling

> ✓ **CHECKPOINT**
> Retrieval and computation are now in your toolkit: LangChain composition, the full RAG pipeline from embedding to injection, and tool calling. The shipped Resume Matcher proves you can take a RAG application from empty repo to working product.

WEEK 08

YOU SHIP **A working coding agent: tools, sandbox, permission layer**

Harness Engineering

DAYS 50–56

The model is not the product; the system around it is.

- What a harness is & harness design principles
- Coding agent I: environment & tools
- Coding agent II: the context assembler
- Coding agent III: policy, safety & humans
- Harness evaluation & failure taxonomy

Stack: Tool sandboxes · permission layers · harness evals

> ✓ **CHECKPOINT**
> You can now design the system around the model: a coding agent with its environment, tools, context assembler, policy layer and human controls, plus an evaluation and failure taxonomy. The harness, not the model, is what you engineer from here.

Weeks 09 & 10

PHASE 4 · MCP & LOOPS

WEEK 09

YOU SHIP **Your coding agent, extended with live MCP servers**

MCP: Model Context Protocol

DAYS 57–63

The open standard that turns your harness into an ecosystem.

- Why a protocol at all
- MCP client & MCP server: both sides of the wire
- MCP in practice: real servers, real tools
- Extending your coding agent with MCP
- MCP integration lab

Stack: MCP clients & servers · filesystem, web & spreadsheet tools

> ✓ **CHECKPOINT**
> Your agent now speaks the industry's open standard: you understand MCP clients and servers on both sides of the wire, and your coding agent operates live MCP servers for real tools. What you built is no longer an island; it is part of an ecosystem.

WEEK 10

YOU SHIP **A self-correcting agent loop with budgets and brakes**

Loop Engineering

DAYS 64–70

A harness that runs once is a function; in a controlled loop, it is an agent.

- The agentic loop
- Loop control: termination, budgets, brakes
- Verification & self-correction loops
- Loop failure modes & hardening lab
- The failure modes that bankrupt teams

Stack: Loop budgets · brakes · verification & tracing

> ✓ **CHECKPOINT**
> Autonomy is now under control: the agentic loop with termination, budgets and brakes, verification and self-correction, and the failure modes that bankrupt teams caught before they run. Your agent corrects itself inside limits you designed.

Weeks 11 & 12

PHASE 5 · GRAPHS & CERTIFICATION

WEEK 11

YOU SHIP **Your agent as a typed, checkpointed LangGraph graph**

Graph Engineering & LangGraph

DAYS 71–77

Loops buy autonomy at the cost of control; graphs buy the control back.

- Why graphs: from loops to topologies
- LangGraph fundamentals & persistence
- Human-in-the-loop & checkpointing
- Graph topologies
- Lab: your agent rebuilt as a state machine

Stack: LangGraph · checkpoints · human gates · time travel

> ✓ **CHECKPOINT**
> Your agent is now a typed, checkpointed LangGraph state machine with human gates and time travel. You can choose a topology deliberately, explain why graphs buy control back from loops, and rebuild an existing agent into one.

WEEK 12

YOU SHIP **YOUR CLAUDE AI CERTIFICATION: earned, not attended**

Claude AI Certification

DAYS 78–84

A full week on the Claude ecosystem, ending in a certification you earn.

- Claude Projects, Artifacts & Claude Code
- Building with the Messages API
- Constitutional AI & responsible use
- Certification preparation
- The build project & the assessment

Stack: Claude ecosystem · Anthropic certification program

> ✓ **CHECKPOINT**
> The Claude ecosystem is now yours end to end: Projects, Artifacts, Claude Code, the Messages API and Constitutional AI, closed out by the certification build project and assessment. The certification on your profile was earned in a build, not attended.

Weeks 13 & 14

PHASE 6 · GPU & CAPSTONE BUILDS

WEEK 13

YOU SHIP **A fine-tuned model on real hardware, deployed to the cloud**

GPU Lab & Cloud Deployment

DAYS 85–91

Fine-tune on real hardware, then deploy on the big three clouds.

- LoRA/PEFT fine-tuning on the RTX 5090
- Fine-tune vs prompt vs RAG: when each wins
- Deploying on Azure OpenAI
- Deploying on AWS Bedrock & Vertex AI
- Cost, keys & monitoring in production

Stack: NVIDIA RTX 5090 · LoRA/PEFT · Azure · Bedrock · Vertex AI

> ✓ **CHECKPOINT**
> You have fine-tuned a real model on the RTX 5090 with LoRA/PEFT and deployed on Azure OpenAI, AWS Bedrock and Vertex AI. You can now argue, with evidence, when to fine-tune, when to prompt and when to RAG, and what each costs to run.

WEEK 14

YOU SHIP **A working Enterprise Knowledge Assistant, ready to ship**

Capstone I: Enterprise Knowledge Assistant

DAYS 92–98

Your production RAG capstone begins: a real system, not a demo.

- Problem definition, goals & system design
- RAG architecture & guardrails
- The data pipeline
- The RAG pipeline
- The application layer

Stack: Your full RAG stack · mentor build reviews

> ✓ **CHECKPOINT**
> Capstone I stands: a production Enterprise Knowledge Assistant with its data pipeline, RAG pipeline, guardrails and application layer designed and built by you, reviewed with your mentor and ready for deployment in the final week.

Weeks 15 & 16

PHASE 7 · SHIP & LAUNCH

WEEK 15

YOU SHIP A working multi-agent support system, ready to ship

Capstone II: Multi-Agent Support System

DAYS 99–105

Your multi-agent capstone: orchestrated, checkpointed and human-gated.

- Problem definition & multi-agent design
- The data & tooling layer
- Agent orchestration in LangGraph
- The application layer
- Checkpoints & human gates in place

Stack: LangGraph orchestration · multi-agent tracing

> ✓ **CHECKPOINT**
> Capstone II stands: a multi-agent support system orchestrated in LangGraph, with its data and tooling layers, checkpoints and human gates in place. Two production-grade systems now carry your name into the final week.

WEEK 16

YOU SHIP CAPSTONES SHIPPED + your career launch kit

Ship, Operate & Career Launch

DAYS 106–112

Both capstones shipped and operated, and your career launched with them.

- CI/CD & evaluation gates for both capstones
- Governance, cost models & tracing
- 30-project GitHub portfolio: wrap & polish
- AI résumé, LinkedIn & interview prep
- Demo Day, graduation & certificates

Stack: CI/CD · evals · observability · your portfolio

> ✓ **CHECKPOINT**
> Both capstones are live behind CI/CD with evaluation gates and observability; your 30-project portfolio is polished; your resume, LinkedIn and interview preparation are done. Demo Day closes the program with your work presented and your career launch under way.

BEYOND THE 16 WEEKS

Use Cases, Certifications & Your Resources

YOUR RESOURCE LINKS

Skool Community	Class recordings, announcements and the student community: join on day one	Join Skool →
Real-Time Use Cases	The industry use-case document you build from throughout the program	Open Use Cases →
Use-Case Tracker (Excel)	Log your progress on every use case as you complete it	Open Tracker →
Anthropic Certifications Training	The Claude certification catalog and how to prepare for each program	Open Guide →
30 GitHub Projects: Assignments	Your assignment library: beginner to advanced portfolio projects	Open Projects →
Technology Certification Encyclopedia	10 hours: verified certification programs across providers, mapped to roles	Open Encyclopedia →

CERTIFICATIONS & TECHNOLOGY WE TRAIN YOU ON

Anthropic: Claude with Anthropic (Claude 101) · Building with the Claude API · Claude Code in Action · Introduction to MCP · Claude on AWS Bedrock · Claude on Google Vertex AI · partner certification pathways. **Plus** the Technology Certification Encyclopedia: a 10-hour guided tour of verified programs across providers, mapped to roles.

14 ENTERPRISE USE CASES WE TEACH ACROSS THE PROGRAM

- ✓ AI Customer Support Chatbot
- ✓ Healthcare Document Summarization
- ✓ Insurance Claims Processing
- ✓ Personalized AI Tutoring
- ✓ Supply Chain Intelligence
- ✓ HR & Recruiting Assistant
- ✓ Sales Call Intelligence
- ✓ Resume Screening & Matching
- ✓ Financial Risk Analysis
- ✓ Retail Recommendation Engine
- ✓ Legal Contract Review
- ✓ Fraud Detection & Alerts
- ✓ Enterprise Knowledge Assistant
- ✓ AI Research Assistant

IN PARALLEL WITH THE PROGRAM

Additional & Optional Classes

These sessions run in parallel with the scheduled classes; they are not counted within them. Optional, but strongly worth attending: each one connects what you are building to the people, tools and interviews waiting on the other side of the program.

Guest Lectures from AI Leaders

Practitioner sessions from industry leaders, slotted between the modules of the program, typically after each capstone, connecting your builds to real teams.

Using Claude Code Effectively

A hands-on session on agentic coding tools, best paired with harness engineering, where the harness you are building is mirrored in a production tool.

No-Code and Low-Code AI Tooling

A guided survey of the no-code and low-code landscape, showing where visual tooling fits alongside engineered systems and when each approach truly wins.

Interview and Portfolio Clinic

Packaging your two capstones into a portfolio that interviews well, plus system-design interview practice focused on real-world agent architectures.

Free Bootcamps & FDE Sessions

FREE bootcamps every two months, plus Forward Deployment Engineering sessions that introduce one of the fastest-growing career paths in applied AI.

Career Guidance & Interview Prep

Ongoing career guidance and interview preparation sessions that continue throughout your batch and beyond it, all the way into your first AI role.

WORK WITH US

KKRGENAI INNOVATIONS

Applied AI training, projects and consulting.

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